MILITARY SPECIFICATION SHEET

BATTERY MAGNESIUM, DRY BA-4386/PRC-25

This specification sheet is approved for use within Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the dry magnesium batteries herein shall consist of this document and the latest issue of MIL-B-55252(CR).

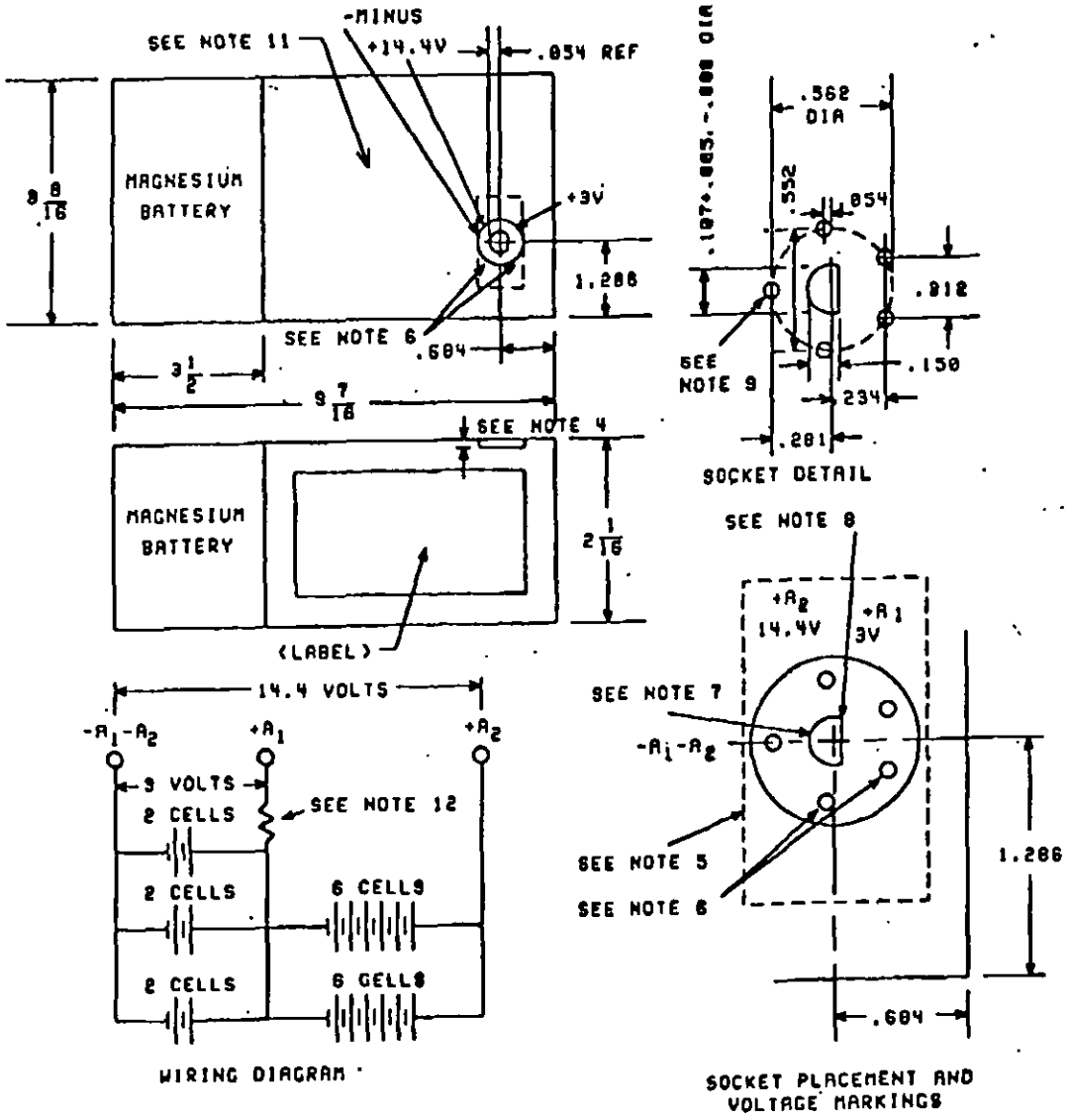


Figure 1. Battery dimensions and wiring diagram

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NOTES:

1. All dimensions are in inches.

2. Unless otherwise specified, all tolerances are ± 0.005 inch for decimals.

3. Tolerances of length, width and height of battery are ± 1/16 inch for fractions.

4. The minimum depth of socket well shall be 1/2 inch.

5. The size of the insulating plate is optional.

6. Socket insert contact not required.

7. Pin-circle center.

8. 'D' hole for plug positioning.

9. Five holes to fit 1/8 in. pins, 7/16 inch in length.

10. Socket must float 3/32 + 1/32 in all directions from the specified pin circle center location in the -1/16 plane of the socket from reference dimension.

11. A printed socket facsimile may be included adjacent to the socket, on the socket side of the battery, in lieu of terminal markings shown in the top view of Fig. 1. The facsimile shall be permanent, legible, include outline of socket, terminal identification shown in aforementioned top view and meet general battery marking requirements covered or referenced herein.

12. A high resistance wire or a dropping resistor may be furnished in this battery and shall be located in the positive leg of the 3 volt unit (A_1) to maintain the A_1 unit maximum closed circuit voltage of 3.0 volts under all capacity discharge conditions of the applicable specifications. If a voltage dropping resistor is furnished, it shall be type RW-69 in accordance with the latest issue of Military Specification MIL-R-26.

13. On all four (4) sides of the battery 'MAGNESIUM BATTERY' shall be printed in 1/2 inch (approx.) bold lusterless green lettering (same as jacket color) on a black background extending a distance of 3 1/2 inches from end of battery.

14. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractural testing.

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REQUIREMENTS:

Dimensions and configuration: See Figure 1.

	A_1 unit	A_2 unit
Nominal voltage	3.0	14.4
Usual number and type of cells:	6 long 'C' cells	18 long 'C' cells
Usual cell connections: (6 ea. cells common to both sections)	Series-parallel	Series-Parallel

Terminals: 5 hole socket (See figure) (no obstruction of any holes)

Weight (maximum): 3.0 pounds

Capacity tests: When the battery is tested in accordance with the methods of examination and test of this specification, the minimum capacity-test requirements shall be as specified below.

Capacity Tests: (Spec. MIL-B-55252A(EL)

Specification Para.	Test Identification	Requirement
4.7.9.1.1	I	55 hours
4.7.9.1.4	See 4.7.9.1.9 & 4.7.9.1.10 below	N/A
4.7.9.1.5	T *	52 hours
4.7.9.1.6	D **	50 hours
4.7.9.1.7 (added req't)	HT(3) *	52 hours
4.7.9.1.8 (added req't)	HT(4) *	55 hours
4.7.9.1.9 (added engrg req't)	LT(1) *	Eng Info (run to end voltage)
4.7.9.1.10 (added engrg req't)	LT(2) *	Eng Info (run to end voltage)

* Discharge A_2 section only

** 52 weeks storage at 113°F only

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Delayed Voltage Test:

a. Tests - HT(3) & HT(4)

(1) HT(3): When the battery is tested in accordance with the methods of examination and test of this specification, the maximum voltage-delay time shall be 15 seconds on the initial discharge cycle and 1.5 seconds on all subsequent cycles to and including the 156 cycle and 180th cycle for engineering information.

(2) HT(4): When battery is tested in accordance with the methods of examination and test of this specification, the maximum voltage-delay time shall be 15 seconds on the initial discharge cycle and 1.5 seconds on all subsequent cycles to and including the 165th cycle and 180th cycle for engineering information.

b. Tests -LT(1) & LT(2) - for engineering info

When the battery is tested in accordance with the methods of examination and test of this specification, the maximum voltage-delay time shall be 15 seconds on the initial discharge cycle and 1.5 seconds on all subsequent cycles to and including the 135th cycle.

Drop test: When the battery is tested in accordance with this specification, the socket shall not move beyond the limits specified herein nor shall the components shift within the jacket, or show evidence of defects that will affect performance, life or serviceability.

First Article Inspection:

Vibration Test
Mechanical Shock Test
Insulation Resistance Test
Battery Venting Test
Visual and Mechanical (External and Internal)
Battery Voltage
Socket Breakage Test
Dimensions & Weight
Drop Test
Waterproofing-of-Jacket Test
Direct Short Test
Corrosion Test
Capacity
I, HT(3)
Delayed Voltage
HT(3)

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Quality Conformance Inspection:

Visual and Mechanical (External)
Battery Voltage
Dimensions & Weight
Drop Test
Waterproofing-of-Jacket Test
Insulation Resistance Test
Battery Venting Test
Direct Short Test
Socket Breakage Test
Corrosion Test
Vibration Test
Mechanical Shock Test
Capacity

T, D, HT(3), HT(4), LT(1)* & LT(2)*

Delayed Voltage

HT(3), HT(4), LT(1)* & LT(2)*

* For Engineering Info

Methods of examination and tests:

Capacity tests: See requirements for capacity specified herein.

Storage -	Service	Test	Period
	55	I	w/o storage
	52	HT(3)	* 4 wks @ 160° F
	55	HT(4)	* 1 wk @ 160° F
	52	T	* 13 wks @ 130° F
	50	D	* 52 wks @ 113° F
	***	LT(1)	** 4 wks @ 160° F
	***	LT(2)	** 1 wk @ 160° F

* Followed by a minimum of 16 hrs at 70° F then 70° F discharge
** Followed by a minimum of 16 hrs at 40° F then discharge at 40° F
*** Engrg info only

Discharge:

a. Tests I,D

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Discharge (Cont'd)

All units shall be discharged concurrently through the resistances listed in Load 1, below, for 2 minutes, then through resistances in Load 2, below, for 18 minutes. This cycle shall be repeated continuously to the applicable test-end voltage listed below.

	A_1 unit	A_2 unit
Resistance:		
Load 1	6.76 ohms	14.2 ohms
Load 2	open circuit	291 ohms
Maximum voltage:		
Open circuit	---	16.6 volts
Closed Circuit	3.0 volts	---
Test end Voltage:	2.12 volts	10.0 volts

b. Tests HT(3) & HT(4), LT(1) & LT(2), T

The A_2 unit shall be discharged through the resistance listed in Load 1, below, for 2 minutes, then through the resistance in Load 2, below, for 18 minutes. This cycle shall be repeated continuously to the applicable test-end voltage listed below.

Resistance	A_1 unit	A_2 unit
Load 1	Open circuit	14.2 ohms
Load 2	Open circuit	291.0 ohms
Maximum voltage:		
Open circuit	---	16.6 volts

Test end voltage:

Delayed voltage test. Voltage delay readings shall be taken on each unit of the battery during capacity discharge in accordance with the following schedule.

a. Test HT(3)

Initial and second discharge cycles
155th and 156th discharge cycles for 51-2/3 to 52d hr requirement
178th and 180th discharge cycles for the 59-2/3 to 60th hour for engineering information

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b. Test HT(4)

Initial and second discharge cycles
164th and 165th discharge cycles for 54-2/3 to 55th hr requirement
179th and 180th discharge cycles for the 59-2/3 to 60th hour for engineering information.

c. Tests LT(1) & LT(2)

Initial and second discharge cycles
134th and 135th discharge cycles (44-2/3 and 45th hour of discharge)

On the first discharge (when Load 1 is applied), the A_2 unit must be above end voltage within 15 seconds after the Load is applied. On all subsequent cycles the A_2 unit must be above end voltage within 1.5 seconds after Load 1 is applied to the A_2 section for that discharge cycle.

BATTERY VENTING TEST

Test Procedure

The battery's A_2 section (14.4 volts nominal) shall be subjected to a 5 ohm discharge for 2 minutes followed by a discharge through 100 ohms for four minutes. This duty cycle shall be repeated continuously until the battery voltage reaches ten (10) volts.

Failure

A failure is defined as any battery exhibiting any dimensional change in excess of 1/16 inch.

First Article Samples

Ten (10) First Article sample batteries shall be subjected to the above described test and no failures shall be permitted.

Noncompliance (First Article)

If one or more sample batteries fail to meet the First Article Battery Venting Test, the supplies shall immediately make the remedial changes and at no additional cost to the government shall be required to submit ten additional First Article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the First Article Inspection report.

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BATTERY VENTING TEST (Cont'd)

Quality Conformance Inspection

(1) Sampling Plan: Sampling for the inspection lot shall conform to special procedures for small-sample inspection of MIL-STD-105 with an S-2 inspection level at an AQL of 1.0% defective. This inspection shall be performed on separate samples selected from the inspection lot and shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

(2) Noncompliance: If the number of defectives for the group of samples selected from the inspection lot equals or exceeds the rejection number for the designated AQL, when tested in accordance with the above described Battery Venting Test, the lot shall be rejected. Acceptance of the monthly shipment lot shall be withheld pending results of corrective action and determination by the government of the acceptability of the corrective action taken.

DIRECT SHORT TEST

Test Procedure:

The battery shall be subjected to a direct short, i.e. less than 0.30 ohms for two hours.

Failure:

A failure is defined as any battery having pieces or parts expelled or ejected from it.

First Article Samples:

Ten (10) First Article sample batteries shall be subjected to the above described test and no failures shall be permitted.

Noncompliance (First Article)

If one or more sample batteries fail to meet the Direct Short Test, the supplier shall immediately make the remedial changes and at no additional cost to the government shall be required to submit ten additional First Article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the First Article Inspection report.

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DIRECT SHORT TEST (Cont'd)

Quality Conformance Inspection

(1) Sampling plan: Ten (10) samples each shall be taken at random from the first five hundred (500) batteries produced in each monthly shipment lot.

(2) Noncompliance: More than one (1) failure per sample group of ten constitutes a failure of the monthly production lot and corrective action must be taken; such as determining cause, reporting same to government, obtaining permission from government to reinitiate production and repeating monthly production lot evaluation.

SOCKET BREAKAGE TEST

Test Procedure:

The socket plate in the battery shall be subjected to a force of twenty (20) pounds by a 1/4 inch diameter rod at any point on the top surface of the battery socket plate.

Failure:

A failure is defined as any socket plate which breaks or drops more than 1/16 inch below the top surface of the battery.

First Article Samples:

Ten (10) First Article sample batteries shall be subjected to the above described test and no failure shall be permitted.

Noncompliance (First Article)

If one or more sample batteries fail to meet the First Article Socket Breakage Test, the supplier shall immediately make the remedial changes and at no additional cost to the government shall be required to submit ten additional First Article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the First Article Inspection report.

Quality Conformance Inspection

(1) Sampling Plan: Sampling for the inspection lot shall conform to special procedures for small sample inspection of MIL-STD-105 with an S-2 inspection level at an AQL of 1.0% defective. This inspection shall be performed on separate samples selected from the inspection lot and shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

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SOCKET BREAKAGE TEST (Cont'd)

(2) Noncompliance: If the number of defectives for the group of samples selected from the inspection lot equals or exceeds the rejection number for the designated AQL, when tested in accordance with the above described Socket Breakage test, the lot shall be rejected. Acceptance of the monthly shipment lot shall be withheld pending results of corrective action and determination by the government of the acceptability of the corrective action taken.

DROP TEST

When performed during First Article Inspection, five (5) each Group II Test Samples shall be subjected to the drop test prior to internal visual and mechanical inspection and waterproofing-of-jacket (WOJ) test. When performed during Acceptance Inspection, five (5) each group B, WOJ Test Samples shall be subjected to the drop test prior to the WOJ test. Each battery shall be dropped once from a height of 30±2 inches onto a hard surface consisting of concrete, wood, or steel. The 3 9/16" x 2 1/16" side of the battery nearest the socket shall be parallel to this surface upon release, and shall make contact with it. If any failure occurs, the contractor shall take immediate action to correct the defect and eliminate the cause and pending this action, quality conformance acceptance shall be withheld. When the nature of the failure has been determined and the necessary correction action taken, the rejected lot and all other batteries in process at the time of the failure shall be reworked to eliminate the defect. Reworked lots shall be reinspected using a sample size of the eighty batteries with no failures permitted.

CELL CORROSION TEST

Test Procedure:

The eighty (80) cells selected for test shall be stored at 130°F for twelve (12) weeks after each is given a serial number. Then a visual inspection of the group of 80 cells shall be made weekly during the 12 weeks storage. In addition, four (4) cells from the group shall be drawn at random every two weeks and inspected internally.

Failure:

A failure is defined as any cell which shows evidence of corrosion, i.e., a mass of white powder. The evidence of corrosion can be on the inner or outer wall of the cell can.

First Article Samples:

Eighty (80) cells selected from the group to be made into First Article sample batteries shall be subjected to the above described test and no failures shall be permitted.

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CELL CORROSION TEST (Cont'd)

Noncompliance (First Article)

If one or more cells shows evidence of corrosion as defined above, the supplier shall immediately make the remedial changes and at no additional cost to the government shall be required to submit eighty (80) additional cells for reinspection. A description of the corrective action taken or to be taken shall be included in the First Article inspection report.

Quality Conformance Inspection. (Group C, Subgroup II)

(1) Sampling Plan. Each monthly shipment lot shall be subject to random sampling of eighty (80) cells from the group to be made into batteries for Group A, B_1, B_2, and C inspection procedures. These eighty (80) cells shall be subjected to the above described test and the acceptance number shall be one (1).

(2) Noncompliance. If the number of cells showing evidence of corrosion exceeds the above designated acceptance number (1), the government shall have cause to refuse to accept subsequent shipment lots until it has been proven, to the satisfaction of the government, that faults revealed by the test have been corrected.

Closed circuit voltage. A direct current voltmeter of proper range and sensitivity as specified in Para. 4.1.1.1 of MIL-B-55252A(EL) shall be used to measure the closed circuit voltage utilizing the load resistance specified below:

UNIT	LOAD RESISTANCE	MINIMUM PERMISSIBLE VOLTAGE (VOLTS)
A_1	6.5 ohms	2.70
A_2	75.0 ohms	12.0

Voltage shall be above "MINIMUM PERMISSIBLE VOLTAGE" within fifteen seconds after load is applied.

Special markings:

On all four (4) sides of battery "MAGNESIUM BATTERY" in 1/2 inch (approx.) bold lusterless green lettering (same as jacket color) shall appear on a black background as shown in figure 1.

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Special markings (Cont'd)

To the right of the battery label and above the socket, print in bold black lettering on the jacket the following identification:

"LONG
MAGNESIUM BATTERY"
LIFE

Height of lettering for "LONG LIFE" shall be one-half that of the lettering for "MAGNESIUM BATTERY".

Unit package, intermediate and shipping containers: Each package or container, in addition to the markings presently required per contract and RFP, shall have the following identification:

"LONG
MAGNESIUM BATTERY"
LIFE

Marking shall be in orange color No. 12246 in accordance with Federal Standard No. 595 and shall be applied by stenciling or through the use of preprinted labels. Marking shall be located on the top and that side of each package or container which denotes the FSN and Battery Type information. Lettering of "LONG LIFE" shall be one-half the size of the lettering for "MAGNESIUM BATTERY".

INSTRUCTIONAL MESSAGE FOR THE BA4386/PRC25 MAGNESIUM BATTERY:

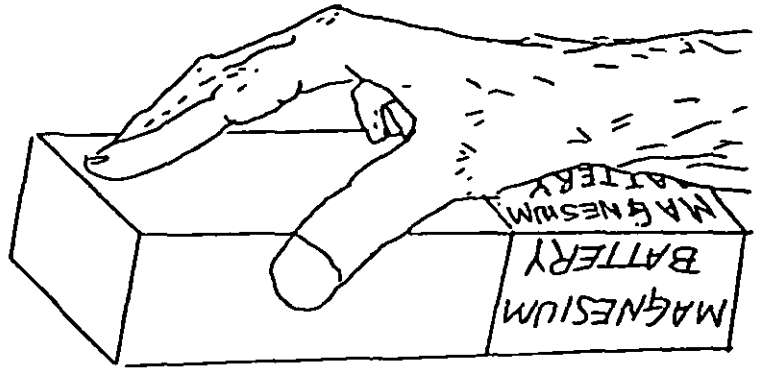
1. An instructional message shall be affixed to the jacket of the BA4386/PRC-25 battery at a location directly opposite the battery socket. The format shall be in accordance with ANSI Z35.I-1972 and shall conform to the following working and graphic:



TO STOP SOCKET BREAKAGE IN AN/PRC-77

WHEN INSTALLING Grip battery at socket end and press gently. Feel Radio Plug Guide Pin enter the battery socket "D" shape guide pin hole until socket mates

WHEN REMOVING Grip battery at socket end and lift straight up.



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2. Standard Colors and Specifications: The background shall be yellow and the panel black with yellow letters. Any letters used against the yellow background shall be black. The colors shall be those of opaque glossy samples as specified in Table I of USA Standard Z53.1-1971.

3. The size of the instructional message shall be consistent with the size of the battery.

4. The size of lettering to be used for the wording of the instructional message should be as large as possible but consistent with good balance and legibility.

5. The coloring and legibility of lettering of the instructional message shall not be affected after the battery is subjected to Para. 4.7.6.2 Waterproofing of Jackets (WOJ) Test of Military Specification MIL-B-55252A(EL), dated 12 April 1971.

6. The instructional message shall be sufficiently durable to meet the requirements of the paragraph entitled 'Permanency and Durability' of MIL-M-13231.

NOTICE: When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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